



DWF_dp_mult_comb_sat function

Combined Unsigned/Signed Multiply and Saturate

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_mult_comb_sat function performs combined (switchable) unsigned/signed multiplication of the two arguments *a* and *b*, truncates the upper bits of the result to the width specified by argument *p_width* and returns a saturated value if an overflow (or underflow) occurs. Argument *a* (*b*) is interpreted as signed if argument *a_tc* (*b_tc*) is 1, otherwise as unsigned. The result must be interpreted as signed if argument *a_tc* or *b_tc* (or both) is 1 (= signed multiplication), otherwise as unsigned (= unsigned multiplication). A dedicated overflow detection (needed for saturation) is used to improve QoR of the multiplier.

Table 1-1 Function Names

Function Name	Description
DWF_dp_mult_comb_sat	VHDL combined multiply and saturate (std_logic_vector/unsigned/signed arguments)
DWF_dp_mult_comb_sat	Verilog combined multiply and saturate
DWF_dp_mult_comb_sat_tc	Verilog combined multiply and saturate (signed arguments)

Table 1-2 Argument Description

Name	Type	Direction	Width / Values	Description
<i>a</i>	Vector	Input	<i>a_width</i>	Input multiplier
<i>a_tc</i>	Bit	Input	1	Two's complement control for multiplier 0 = Unsigned 1 = Signed
<i>b</i>	Vector	Input	<i>b_width</i>	Input multiplicand
<i>b_tc</i>	Bit	Input	1	Two's complement control for multiplicand 0 = Unsigned 1 = Signed
<i>p_width</i>	Integer	Input	≥ 2	Word length of return value (VHDL only, constant)
DWF_dp_mult_comb_sat	Vector	Output	<i>p_width</i>	Returned value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
a_width	≥ 2	Word length of input a
b_width	≥ 2	Word length of input b
p_width	≥ 2	Word length of returned value

Verilog Include File: DW_dp_mult_comb_sat_function.inc

Functional Description

```

z[p_width-1:0] = DWF_dp_mult_comb_sat (a[a_width-1:0], a_tc, b[b_width-1:0], b_tc, p_width)

p[a_width+b_width-1:0] (unsigned) = a (unsigned) * b (unsigned)    if a_tc = 0 and b_tc = 0

p[a_width+b_width-1:0] (signed)  = a (signed)    * b (unsigned)    if a_tc = 1 and b_tc = 0
                                = a (unsigned) * b (signed)      if a_tc = 0 and b_tc = 1
                                = a (signed)    * b (signed)      if a_tc = 1 and b_tc = 1

z[p_width-1:0] = 2p_width-1      if (a_tc = 0 and b_tc = 0) and
                                (p[a_width+b_width-1:0] > 2p_width-1)
                                = 2p_width-1-1    else if (a_tc = 1 or b_tc = 1) and
                                (p[a_width+b_width-1:0] > 2p_width-1-1)
                                = -2p_width-1     else if (a_tc = 1 or b_tc = 1) and
                                (p[a_width+b_width-1:0] < -2p_width-1)
                                = p[p_width-1:0]  else

```

For more information about the DesignWare datapath functions, refer to the topic titled [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_mult_comb_sat_test is
  port (a, b, c      : in  unsigned(7 downto 0);
        a_tc, b_tc  : in  std_logic;
        z           : out unsigned(7 downto 0));
end DWF_dp_mult_comb_sat_test;

architecture rtl of DWF_dp_mult_comb_sat_test is
begin
  z <= DWF_dp_mult_comb_sat (a, a_tc, b, b_tc, 8) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_mult_comb_sat_test (a, a_tc, b, b_tc, c, z);

    input signed [7:0] a, b, c;
    input             a_tc, b_tc;
    output signed [7:0] z;

    // Passes the parameters to the function
    parameter a_width = 8;
    parameter b_width = 8;
    parameter p_width = 8;

    // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
    `include "DW_dp_mult_comb_sat_function.inc"

    assign z = DWF_dp_mult_comb_sat (a, a_tc, b, b_tc) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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